

Supplementary Material for A Comparative Study of Graph Representations for GNN-Based Power Grid Control in L2RPN

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S1. ADDITIONAL EXPERIMENTAL DETAILS

TABLE S-I: Node features for the different graph representations. † Features in the *generator-specific*, *bus-specific*, and *line-specific* groups are encoded in dedicated slots of the 28-dimensional Elements feature vector and are zero-padded for all other node types.

	Default	Heterogeneous	Substation	Elements	PTDF	LODF	ZBus	Sub. + PTDF	Elem. + LODF	Sub. + ZBus
<i>Electrical state</i>										
Active power p	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Reactive power q	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Voltage magnitude v	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Voltage angle $\cos \theta$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Voltage angle $\sin \theta$	✓	✓	✓	✗	✓	✓	✓	✓	✗	✓
Absolute active power $ p $	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
Line loading ρ	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
<i>Forecast</i>										
Active power forecast $\hat{p}$	✓	✓	✓	✓	✓	✗	✓	✓	✓	✓
Reactive power forecast $\hat{q}$	✓	✓	✓	✓	✓	✗	✓	✓	✓	✓
<i>Generator-specific</i> †										
Normalized generation	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
Ramp up/down limits	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
Up/downtime limits	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
Generation costs	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
Generator type (one-hot $\times 5$)	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
<i>Bus-specific</i> †										
Bus type (one-hot)	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
Substation cooldown	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
<i>Line-specific</i> †										
Timestep overflow	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
Time before line cooldown	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
Next maintenance duration	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
Line status	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗

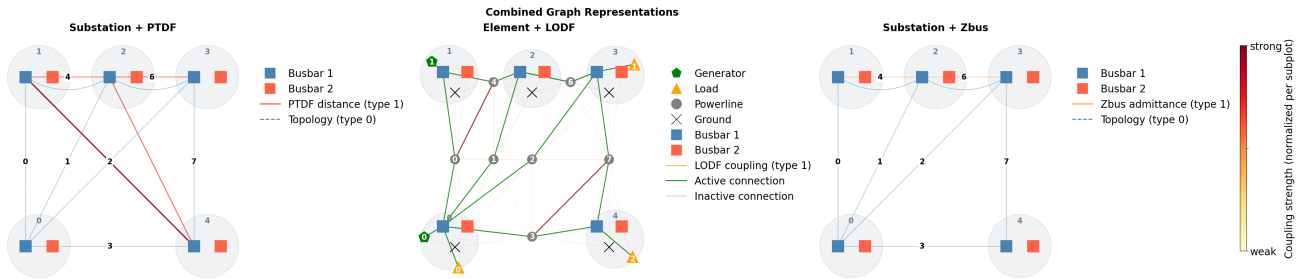


Fig. S-1: Hybrid representations of the *rte_case5_example* environment.

TABLE S-II: GNN Architecture

Parameter	Default	Description
hidden_dim	256	Hidden dimension per GNN layer
out_dim	64	Output embedding dimension
num_layers	3	Number of message-passing layers
fcnet_hiddens	[256, 256, 256]	Policy/value FC layers
fcnet_activation	relu	Activation function
post_fcnet_hiddens	[256, 128]	Layers between MLP output and policy head

TABLE S-III: PPO Hyperparameters

Parameter	Default	Description
lr	1e-4	Learning rate
gamma	0.99	Discount factor
lambda	0.95	GAE lambda
clip_param	0.3	PPO clip epsilon
vf_loss_coeff	0.5	Value loss weight
entropy_coeff	0.0	Entropy bonus weight
grad_clip	40.0	Gradient clipping
num_sgd_iter	5	SGD passes per batch
sgd_minibatch_size	256	Minibatch size
train_batch_size	1024	Rollout batch size
vf_clip_param	100	Value function clip range

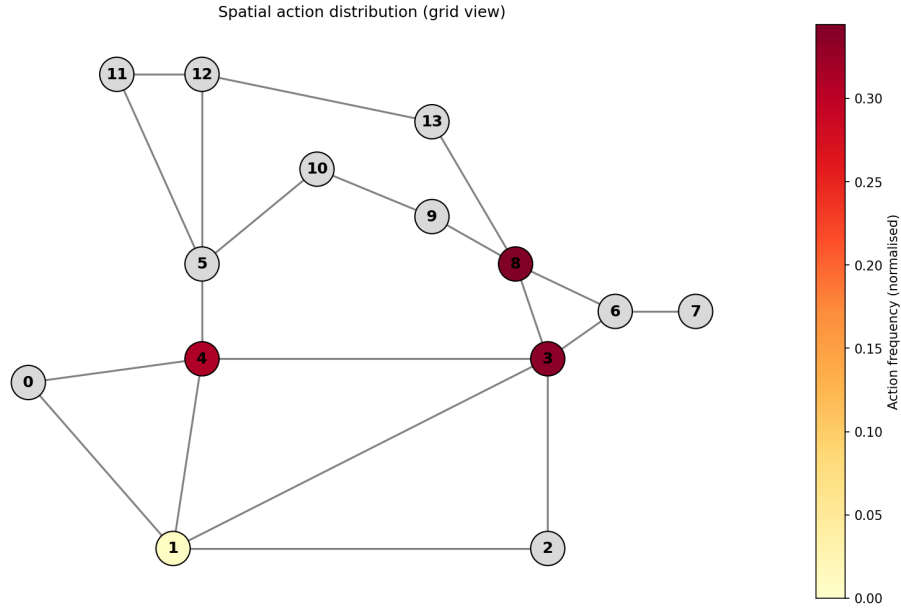


Fig. S-2: All trained agents apply actions predominantly on substations 1,3,4, and 8.

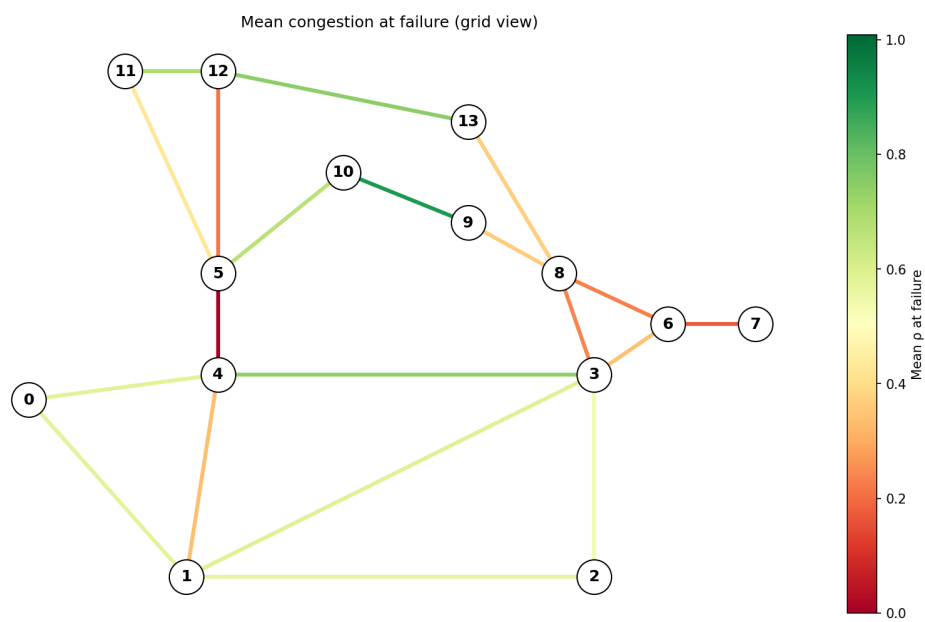


Fig. S-3: Congestion averaged over pre-fault states occur mostly in the 4-5-12 corridor and the 3-8-6-7 region.